

BASIM MATHAKIYA

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EDUCATION

Bachelor of Technology in Computer Science and Engineering

2024 – 2028

Sardar Vallabhbhai National Institute of Technology, Surat

CGPA: **7.04/10**

TECHNICAL SKILLS

Languages: C++, C, Python, JavaScript, MYSQL

Web Technologies: HTML, CSS, React, Vite

Developer Tools: Git, GitHub, VS Code

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks

PROJECTS

Reinforcement Learning Agent for Subway Surfers

Python — PPO — Pygame

- Developed a custom Subway Surfers-inspired game environment in Python using Pygame for reinforcement learning experiments.
- Trained an autonomous agent using Proximal Policy Optimization (PPO) to learn obstacle avoidance and decision making.
- Designed reward functions and tuned training parameters through iterative experimentation.
- Evaluated agent performance over multiple training episodes and refined the policy based on observed behaviour.

Banking Management System

React — JavaScript — HTML — CSS

- Developed a responsive banking application frontend using React and JavaScript.
- Designed reusable UI components for account management and transaction workflows.
- Integrated frontend pages with backend APIs for user authentication and banking operations.
- Focused on creating an intuitive and responsive user experience.

PROBLEM SOLVING

- Solved **400+** algorithmic problems across LeetCode, takeUforward and Codeforces using C++ and Python.
- Strong understanding of Dynamic Programming, Trees, Graphs, Binary Search, Greedy Algorithms, Sliding Window and Backtracking.
- Developed efficient solutions with emphasis on time and space complexity optimization.

RELEVANT COURSEWORK

Data Structures & Algorithms
Database Management Systems

Object-Oriented Programming
Computer Networks